

Architecture summary

One recruiter. One job. Auditable stage changes and read-only search.

Repository: <https://github.com/diyaavirmani/mini-hiring-pipeline>

Request and data flow

01 Browser	02 FastAPI	03 Services	04 SQLite
Same-origin UI and signed login	Authenticated, validated requests	Stage rules and search filters	Profiles and immutable events

Integrity and search boundary

SQLite triggers enforce Applied as the first event, one-step moves, rejection before Hired, terminal outcomes, and immutable history. A serialized write transaction checks the expected stage, so stale concurrent moves fail.

The newest event determines current stage and stage duration. Search combines deterministic filters and typo-tolerant name ranking. Only unresolved queries reach Gemini; its structured filters are validated and executed read-only. AI cannot change candidates or decide transitions.

Scope

One recruiter manages candidates for one job through Applied → Screening → Interview → Offer → Hired. Rejection is allowed before Hired. The browser board supports candidate creation, stage actions, current-stage duration, complete event history, and one search box.

Main trade-offs

SQLite keeps setup simple and supports this one-recruiter workload, but is not intended for horizontally scaled instances or sustained concurrent write traffic. A same-origin interface avoids a separate frontend deployment and cross-origin session setup. The AI fallback makes natural-language searches more flexible but depends on provider availability and adds latency; deterministic searches need no AI credential. The local database is not encrypted at rest.

Verification snapshot

The documented test suite covers transition and audit integrity, concurrent moves, API validation and access control, configuration, search parsing/ranking, AI failure and output validation, and repeat-safe sample seeding. Search evaluation covers all assignment examples plus combined, valid-empty, and unsupported queries. See the repository README for the latest run results and local setup instructions.

Out of scope

AWS and Docker deployment are out of scope for this iteration.

The Markdown file beside this PDF is its editable source. The README contains the latest test and search evaluation results.